
2. Social Network Structure, Measures & Visualization

Contents Network Visualization

- Basics of Social Network Structure - Nodes, Edges & Tie
- Describing the Networks Measures –
 - Degree Distribution, Density,
 - Connectivity,
 - Centralization
 - Tie Strength & Trust



Why do we need Network Measures?

- Who are the central figures (influential individuals) in the network?
 - **Centrality**
- What interaction patterns are common in friends?
 - **Reciprocity and Transitivity**
 - **Balance and Status**
- Who are the like-minded users and how can we find these similar individuals?
 - **Similarity**
- To answer these and similar questions, one first needs to define measures for quantifying **centrality**, **level of interactions**, and **similarity**, among others.

Sample Undirected Network

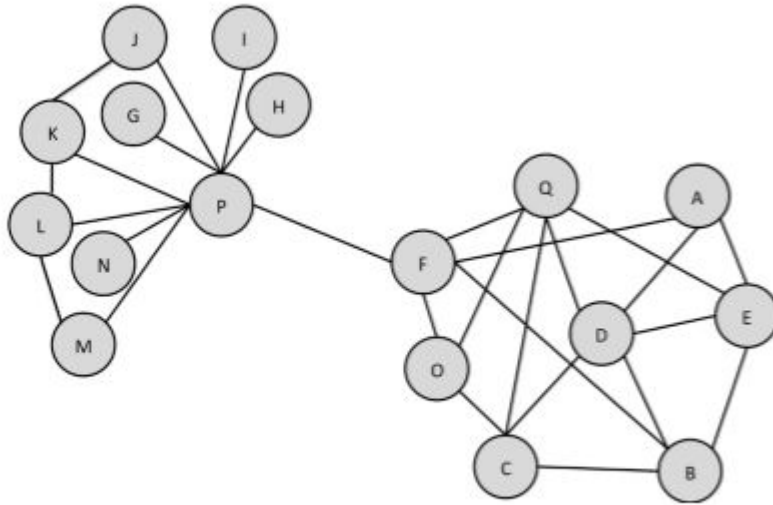


FIGURE 3.1

A sample undirected network.

A Sample Directed Graph

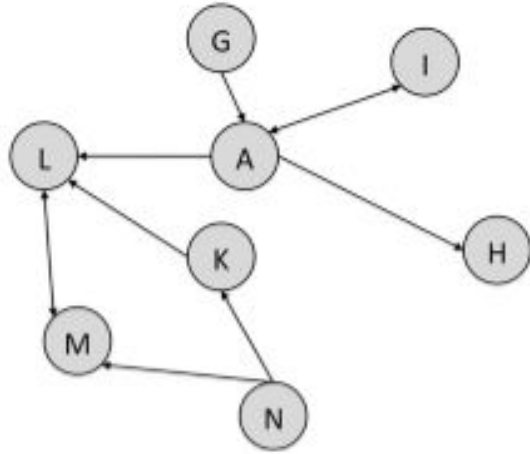


FIGURE 3.2

A sample directed graph.

Centrality

Centrality defines how important a node is within a network

Centrality

- Centrality is one of the core principles of network analysis. It measures **how “central” a node is in the network**. This is used as an estimate of its importance in the network.
- Depending on the application and point of view, what counts as **“central”** may vary **depending on the context**.
- Four ways to compute Centrality-
 - Degree centrality,
 - Closeness centrality
 - Betweenness centrality
 - Eigenvector centrality.

1. Degree Centrality

**Centrality in terms of those
who you are connected to**

Centrality

Degree centrality is a good measure of the total connections a node has, **but will not necessarily indicate the importance of a node** in connecting others or how central it is to the main group.

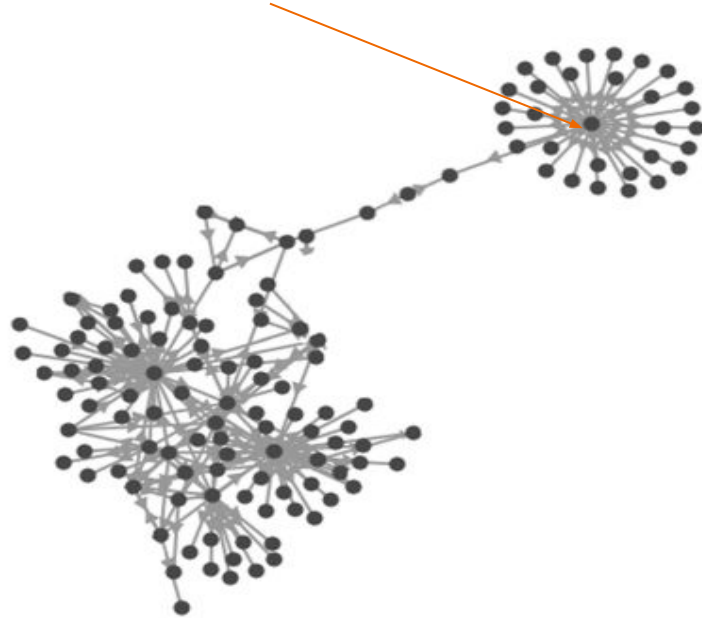


FIGURE 3.3

The node at the center of the cluster in the upper right would have a high degree centrality, even though it is far from the dense center of the network.

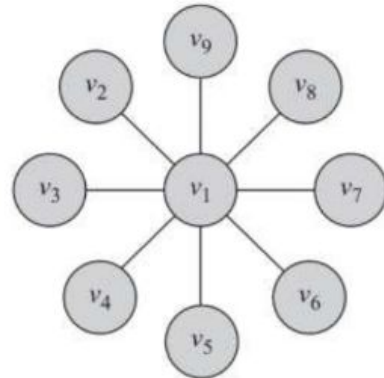
Centrality

- **Degree centrality:** ranks nodes with more connections higher in terms of centrality

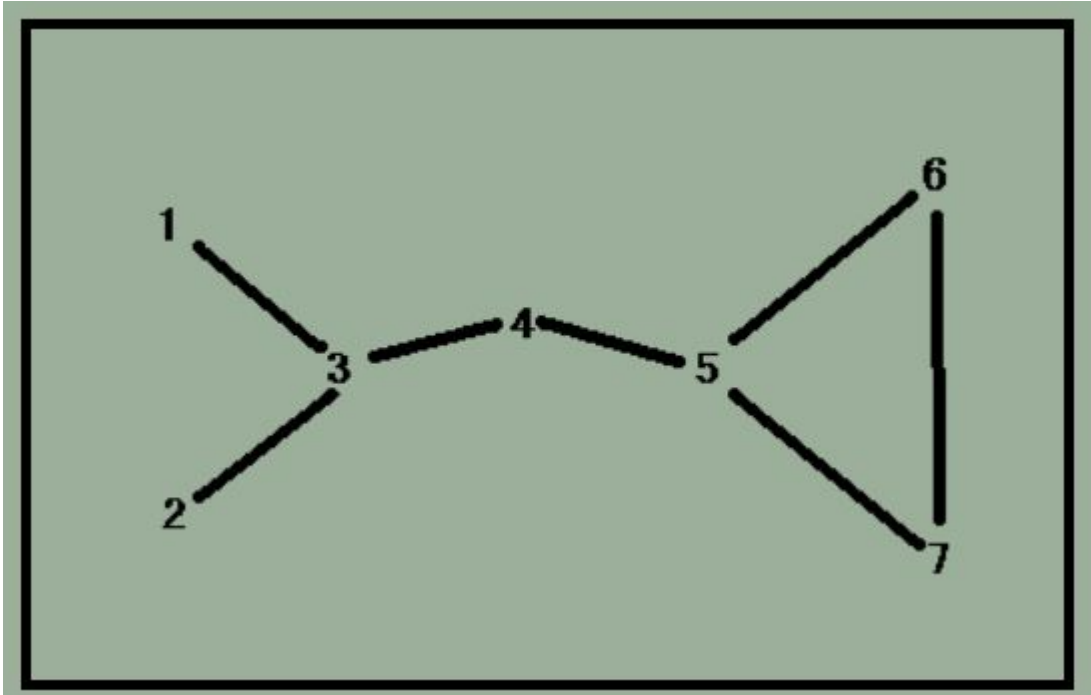
$$C_d(v_i) = d_i$$

- d_i is the degree (number of friends) for node v_i
 - i.e., the number of length-1 paths (can be generalized)

In this graph, degree centrality for node v_1 is $d_1=8$ and for all others is $d_j = 1, j \neq 1$

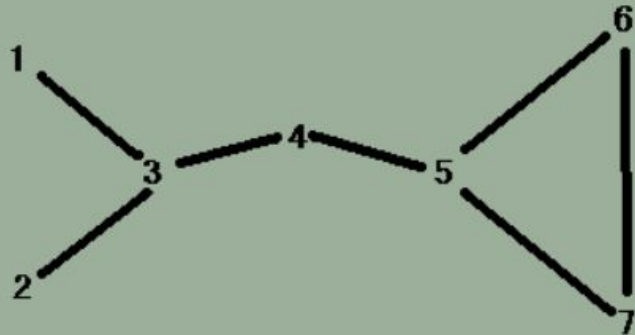


Find degree centrality of each node?



- **Degree centrality** of a node refers to the **number of edges attached to the node**. In order to know the standardized score, you need to **divide each score by $n-1$**

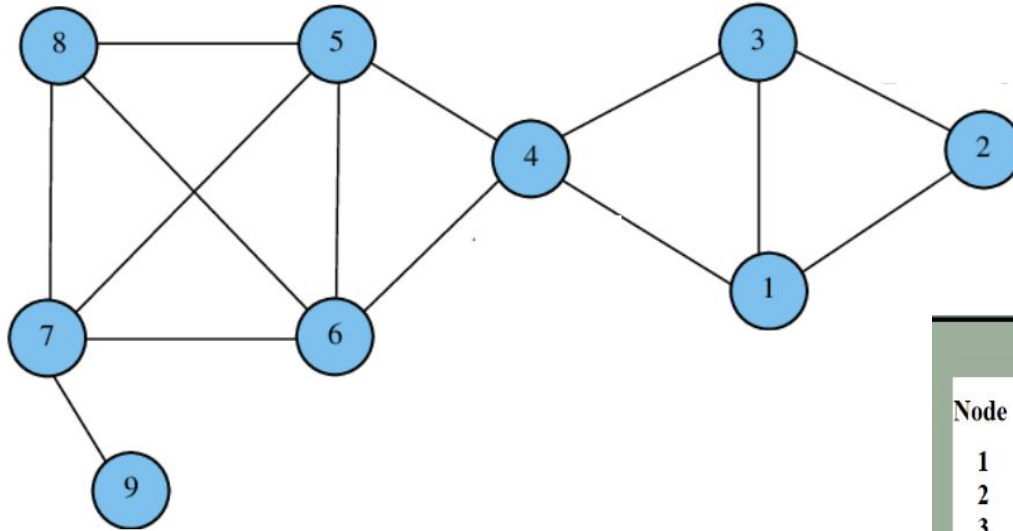
Degree centrality of each node:



Degree Centrality

Node	Score	Standardized Score
1	1	$1/6$
2	1	$1/6$
3	3	$3/6 = 1/2$
4	2	$2/6 = 1/3$
5	3	$3/6 = 1/2$
6	2	$2/6 = 1/3$
7	2	$2/6 = 1/3$

Find degree centrality of each node



Degree Centrality		
Node	Score	Standardized Score
1		
2		
3		
⋮		
⋮		
⋮		
9		

2. Closeness centrality

Closeness centrality

- Closeness centrality indicates **how close a node is to all other nodes** in the network.
- It is calculated as **the average of the shortest path** length from the node to every other node in the network.

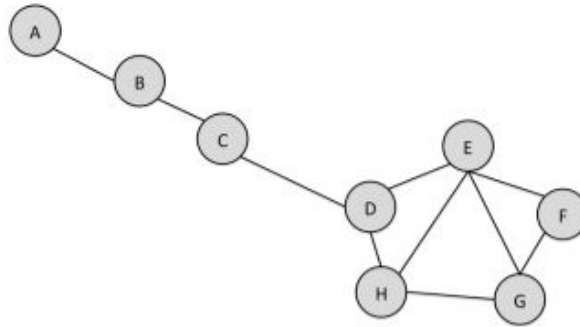


FIGURE 3.4
A sample network.

Closeness centrality

Since node D's closeness centrality is $(3+2+1+1+2+2+1)/7 = 1.71$ and node A's is 3.43, **node D is more central by this**. It indicates **nodes as more central** if they are closer to most of the nodes in the graph **measure**

Table 3.1 The Shortest Path Lengths from D to each Other Node in the Network

Node	Shortest Path from D
A	3 (D → C → B → A)
B	2
C	1
E	1
F	2
G	2
H	1

Table 3.2 The Shortest Path Length from node A to Every Other Node in the Network

Node	Shortest Path from A
B	1
C	2
D	3
E	4
F	5
G	5
H	4

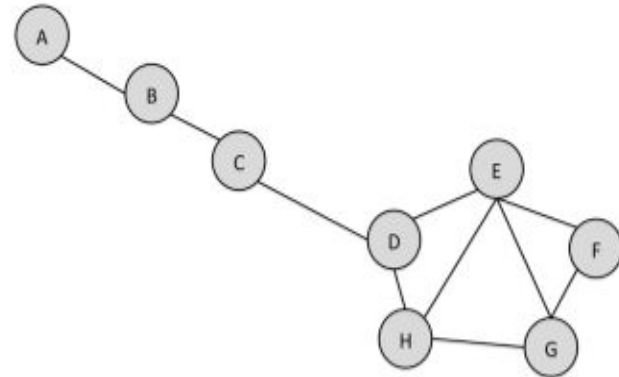
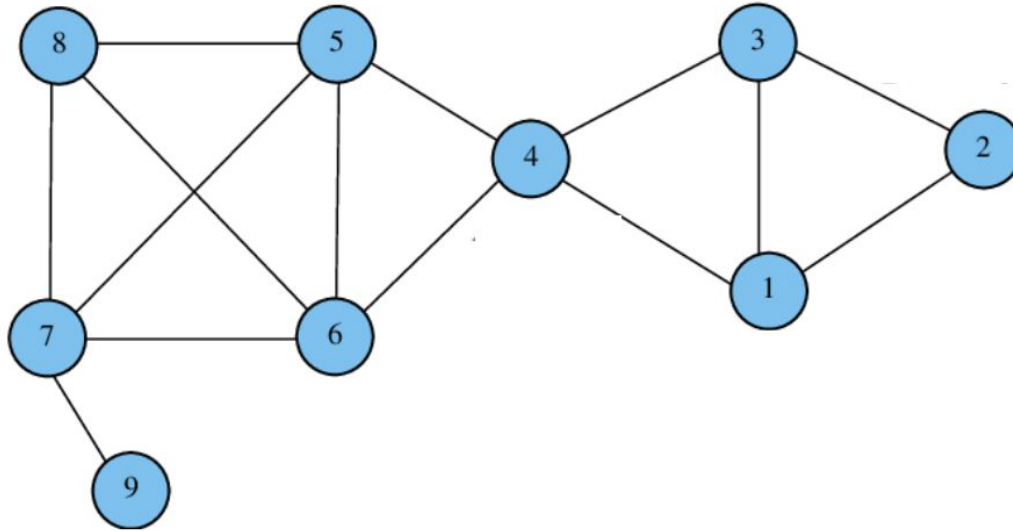


FIGURE 3.4

A sample network.

Find closeness centrality of node 8?



Betweenness centrality

- Measures **how important a node is to the shortest paths** through the network.
- For Calculating Betweenness
 - We select a **pair of nodes** and **find all the shortest paths between those nodes**.
 - We compute the **fraction of those shortest paths that include node N**.

- The vertex betweenness centrality $BC(v)$ of a vertex v is defined as follow

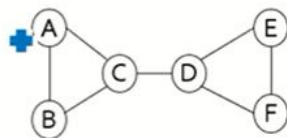
$$BC(v) = \sum_{u,w \in V} \left(\frac{\sigma_{uw}(v)}{\sigma_{uw}} \right)$$

σ_{uw} = Total number of shortest paths between node u and w .

$\sigma_{uw}(v)$ = Total number of shortest paths between node u and w that pass through v .

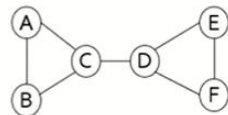


Calculate the between centrality for node C



	σ_{uw}	$\sigma_{uw}(v)$	$\sigma_{uw}(v) / \sigma_{uw}$
(A,B)	1	0	0
(A,D)	1	1	1
(A,E)	1	1	1
(A,F)	1	1	1
(B,D)	1	1	1
(B,E)	1	1	1
(B,F)	1	1	1

Calculate the between centrality for node C



	σ_{uw}	$\sigma_{uw}(v)$	$\sigma_{uw}(v) / \sigma_{uw}$
(D,E)	1	0	0
(D,F)	1	0	0
(E,F)	1	0	0

Betweenness Centrality for C = 6

Betweenness Centrality for B = ?

$$AC = 1/1 = 1$$

$$AD = 1/1 = 1$$

$$AE = 0/1 = 0$$

$$AF = 0/1 = 0$$

$$CD = 0/1 = 0$$

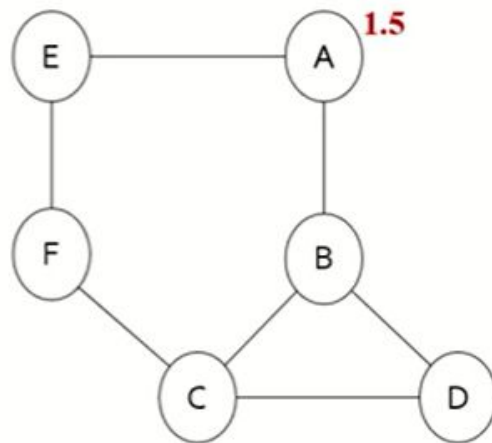
$$CE = 0/1 = 0$$

$$CF = 0/1 = 0$$

$$DE = 1/2 = 0.5$$

$$DF = 0/1 = 0$$

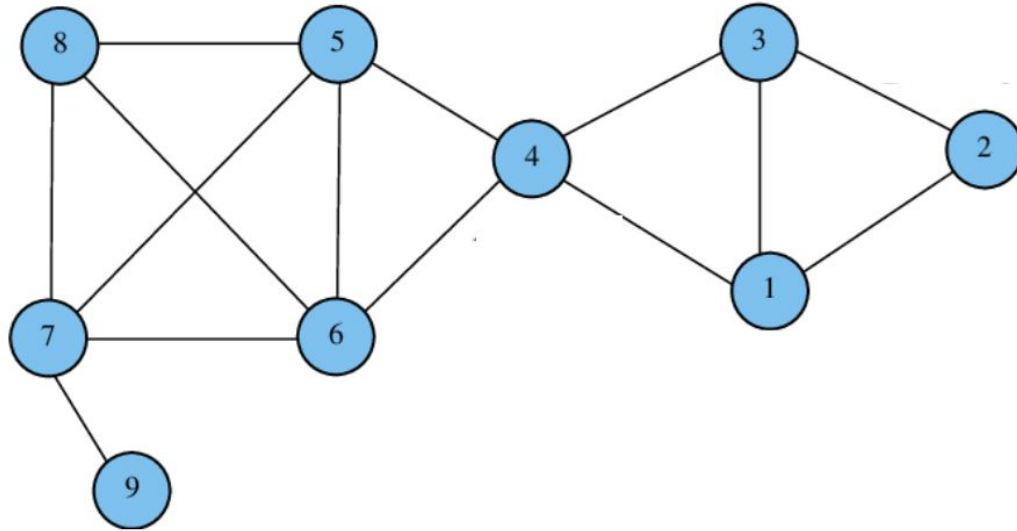
$$EF = 0/1 = 0$$



Betweenness Centrality for B = 2.5



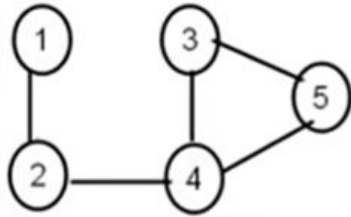
Find Betweenness Centrality of node 4?



Eigenvector centrality

- Eigenvector centrality measures a **node's importance** while giving consideration to the **importance of its neighbors**.
- For example, a node with 300 relatively unpopular friends on Facebook would have lower eigenvector centrality than someone with **300 very popular friends (like Barak Obama)**.
- It is sometimes used to measure a **node's influence** in the network.
- It is determined by performing a **matrix calculation** to determine what is called the *principal eigenvector* using the adjacency matrix.
- Links from important nodes (as measured by degree centrality) are worth more than links from unimportant nodes.

EigenVector Centrality Example (1)



$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

Let $X_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$

Iteration 1

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 3 \\ 2 \end{bmatrix} \equiv \begin{bmatrix} 0.213 \\ 0.426 \\ 0.426 \\ 0.639 \\ 0.426 \end{bmatrix}$$

Normalized Value = 4.69

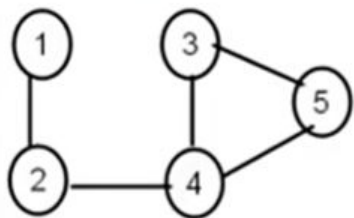
Normalized value = $\text{Sq_root}(1^2 + 2^2 + 2^2 + 3^2 + 2^2)$

Iteration 2

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0.213 \\ 0.426 \\ 0.426 \\ 0.639 \\ 0.426 \end{bmatrix} = \begin{bmatrix} 0.426 \\ 0.852 \\ 1.065 \\ 1.278 \\ 1.065 \end{bmatrix} \equiv \begin{bmatrix} 0.195 \\ 0.389 \\ 0.486 \\ 0.584 \\ 0.486 \end{bmatrix}$$

Normalized Value = 2.19

EigenVector Centrality Example (1)



$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

Let $X_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$

Iteration 3

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0.195 \\ 0.389 \\ 0.486 \\ 0.584 \\ 0.486 \end{bmatrix} = \begin{bmatrix} 0.389 \\ 0.779 \\ 1.07 \\ 1.361 \\ 1.07 \end{bmatrix} \equiv \begin{bmatrix} 0.176 \\ 0.352 \\ 0.484 \\ 0.616 \\ 0.484 \end{bmatrix}$$

Normalized Value = 2.21

Iteration 4

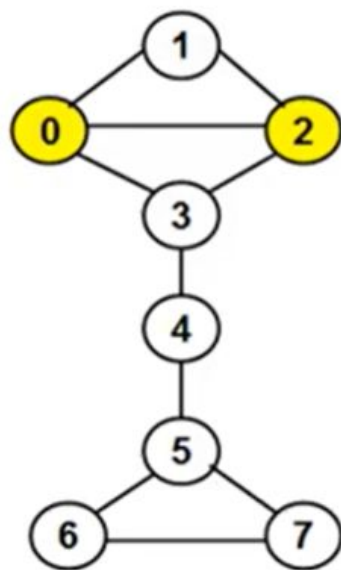
$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0.176 \\ 0.352 \\ 0.484 \\ 0.616 \\ 0.484 \end{bmatrix} = \begin{bmatrix} 0.352 \\ 0.792 \\ 1.100 \\ 1.320 \\ 1.100 \end{bmatrix}$$

Normalized Value = 2.21 converges

Eigen Vector
Centrality

$$\begin{bmatrix} 1 & 0.176 \\ 2 & 0.352 \\ 3 & 0.484 \\ 4 & 0.616 \\ 5 & 0.484 \end{bmatrix}$$

Eigenvector Centrality (1)



Time Complexity: $\Theta(V^3)$

Power Iteration Method

Iteration 1

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 3 \\ 3 \\ 2 \\ 3 \\ 2 \\ 2 \end{bmatrix} \equiv \begin{bmatrix} 0.416 \\ 0.277 \\ 0.416 \\ 0.416 \\ 0.277 \\ 0.416 \\ 0.277 \\ 0.277 \end{bmatrix}$$

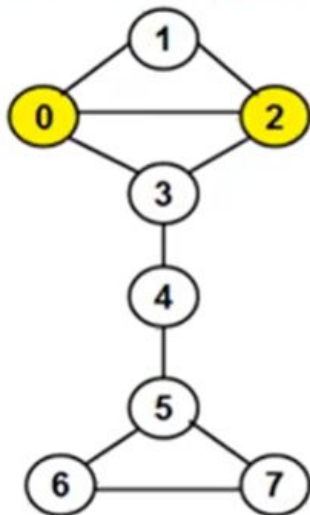
Normalized Value = 7.21

Iteration 2

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 0.416 \\ 0.277 \\ 0.416 \\ 0.416 \\ 0.277 \\ 0.416 \\ 0.277 \\ 0.277 \end{bmatrix} = \begin{bmatrix} 1.109 \\ 0.832 \\ 1.109 \\ 1.109 \\ 0.832 \\ 0.831 \\ 0.693 \\ 0.693 \end{bmatrix} \equiv \begin{bmatrix} 0.428 \\ 0.321 \\ 0.428 \\ 0.428 \\ 0.321 \\ 0.321 \\ 0.268 \\ 0.268 \end{bmatrix}$$

Normalized Value = 2.59

Eigenvector Centrality (2)



Iteration 3

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 0.428 \\ 0.321 \\ 0.428 \\ 0.428 \\ 0.321 \\ 0.321 \\ 0.268 \\ 0.268 \end{bmatrix} = \begin{bmatrix} 1.137 \\ 0.856 \\ 1.284 \\ 1.177 \\ 0.749 \\ 0.857 \\ 0.589 \\ 0.589 \end{bmatrix} \equiv \begin{bmatrix} 0.441 \\ 0.321 \\ 0.481 \\ 0.441 \\ 0.281 \\ 0.321 \\ 0.221 \\ 0.221 \end{bmatrix}$$

Normalized Value = 2.67

Iteration 4

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 0.441 \\ 0.321 \\ 0.481 \\ 0.441 \\ 0.281 \\ 0.321 \\ 0.221 \\ 0.221 \end{bmatrix} = \begin{bmatrix} 1.243 \\ 0.922 \\ 1.203 \\ 1.203 \\ 0.762 \\ 0.723 \\ 0.542 \\ 0.542 \end{bmatrix} \equiv \begin{bmatrix} 0.471 \\ 0.349 \\ 0.456 \\ 0.456 \\ 0.289 \\ 0.274 \\ 0.205 \\ 0.205 \end{bmatrix}$$

Normalized Value = 2.64

After 7 iterations

Vertex ID	0	1	2	3	4	5	6	7
Principal Eigenvector	0.489	0.364	0.489	0.467	0.264	0.232	0.155	0.155

Describing Network

Describing networks - degree distribution

- **Degree distribution** is a way to describe the **distribution of degrees among the nodes in a network.**
- To create a degree distribution, **the degree of each node in the network is calculated.**
- The next step is to **count how many nodes have each degree** and **create a table** or chart to display this information.
- **A bar graph** is a common way to display a degree distribution, with the x-axis showing the degree and the y-axis indicating the number of nodes with that degree.

Describing networks

- Degree distribution

Table 3.3 Degrees for each Node Shown in Figure 3.1

Node	Degree
A	3
B	4
C	4
D	5
E	4
F	4
G	1
H	1
I	1
J	2
K	3
L	3
M	2
N	1
O	2
P	9
Q	5

Table 3.4 The Degree Distribution for the Network in Figure 3.1. The First Column Shows the Degree, and the Second Column Shows How Many Nodes have that Degree

Degree	Number of Nodes
1	4
2	3
3	3
4	4
5	2
6	0
7	0
8	0
9	1

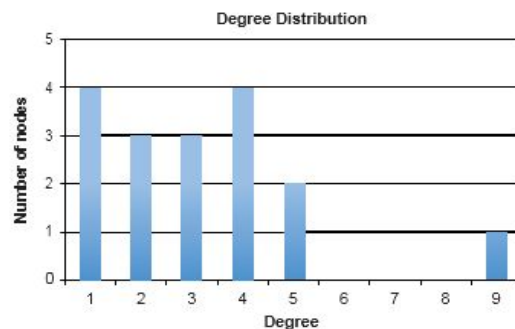
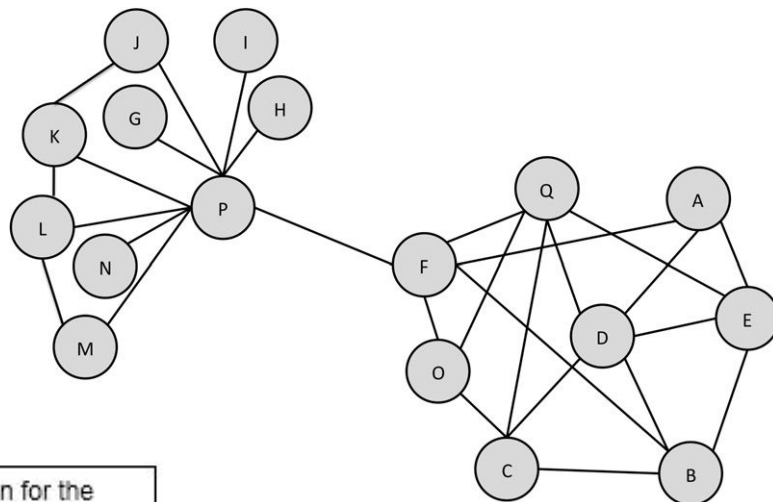


FIGURE 3.5

The degree distribution for the graph shown in Figure 3.1.

Density

- Density describes how connected a network is.
- It is a statistic **comparing the number of edges that exist in a network to the number of edges that could possibly exist.**
- Consider the following two networks, which both have the same number of nodes.
 - Network (a) has very few edges while
 - Network (b) has numerous edges among the same number of nodes.
 - **Network (b) has higher density.**

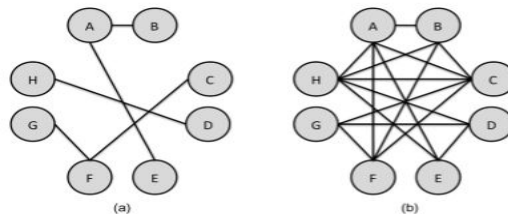


FIGURE 3.6

Network (a) on the left has fewer edges than network (b) on the right. Since they both have the same number of nodes and thus the same number of possible edges, network (b) is more dense.

Density

We can use density to describe a network as a whole. Consider the networks in Figure 3.6. Both have eight nodes. Network (a) has five edges. Since it is an undirected network, the density is $(5/(8*(8-1)/2))$ or $5 \div 28 = 0.179$. Network (b) has 16 edges, so the density is $16 \div 28 = 0.571$. Note that the density is higher for network (b), meaning it's denser.

In an undirected network with n nodes and e edges, the density formula is:

$$\frac{e}{n*(n-1)/2}$$

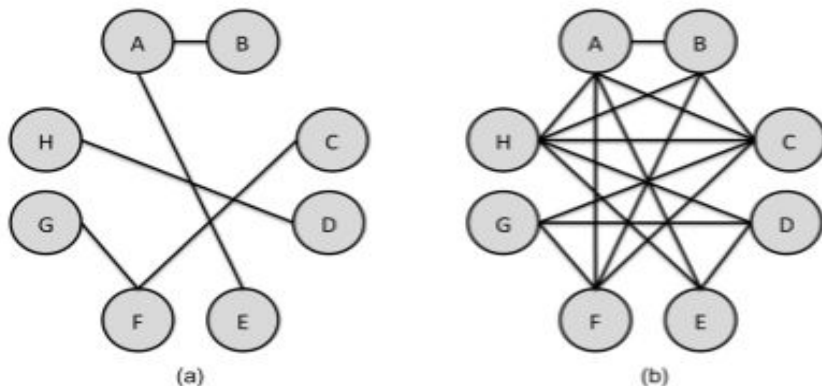


FIGURE 3.6

Network (a) on the left has fewer edges than network (b) on the right. Since they both have the same number of nodes and thus the same number of possible edges, network (b) is more dense.

Density

Now these formulas can be used to calculate density. In a directed network with n nodes and e edges, the formula for density is:

$$\frac{e}{n*(n-1)}$$

In an undirected network with n nodes and e edges, the density formula is:

$$\frac{e}{n*(n-1)/2}$$

Density in egocentric networks/ local clustering coefficient

- we use the **1.5-diameter network**: We consider the node's connections and all the connections between those nodes.
- For this calculation, the **ego-node will be excluded** from its egocentric network because the density of interest is that of the connections between the node's friends.
- **Node A** is connected to nodes **B, E, and H**. To get the **1.5-diameter egocentric network**, we will look at only nodes **B, E, and H** and the connections between them.

Connectivity

Connectivity

The **connectivity** is 1 because removing node B, C, or D would disconnect the graph.

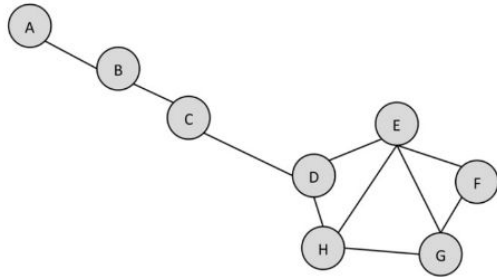


FIGURE 3.4

A sample network.

The **connectivity** is 2

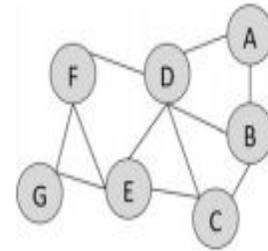


FIGURE 3.8

A sample network with a connectivity of 2.

Centralization

Centralization

- Centralization measures the extent to which the ties of a given network are concentrated on a single actor or group of actors.
- $C(n)$ be the centrality of node n , using whatever centrality measure we choose.
- n^* is the most central node.
- Find the difference in centrality between n^* and every other node in the network, and add those up.

$$\sum_{i=1}^N C(n^*) - C(n_i)$$

Centralization

- n^* is the most central node.
- **Find the difference in centrality** between n^* and **every other node** in the network, and add those up.

$$\sum_{i=1}^N C(n^*) - C(n_i)$$

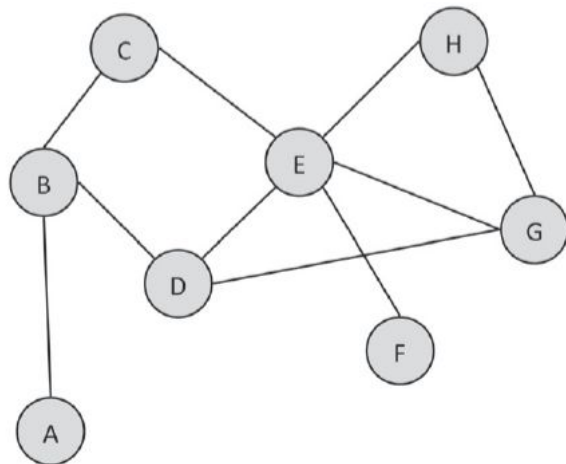
divide this by the **sum of the maximum possible differences** between n^* and **every other node**.

$$\max \sum_{i=1}^N C(n^*) - C(n_i)$$

Centralization

centralization =

$$\frac{\sum_{i=1}^N C(n^*) - C(n_i)}{\max \sum_{i=1}^N C(n^*) - C(n_i)}$$



1. Answer the following questions about this graph.
 - a. What is the degree distribution for this graph?
 - b. What is the density of this graph?
 - c. Draw the 1.5 egocentric network of node G.
 - d. Which node(s) have the highest degree? What is the degree?
 - e. Which node(s) have the lowest degree? What is the degree?
 - f. Which node has the highest closeness centrality?
 - g. Which node has the highest degree centrality?
 - h. What is the centralization (based on degree) of the graph?

Now these formulas can be used to calculate density. In a directed network with n nodes and e edges, the formula for density is:

$$\frac{e}{n*(n-1)}$$

In an undirected network with n nodes and e edges, the density formula is:

$$\frac{e}{n*(n-1)/2}$$

Tie Strength

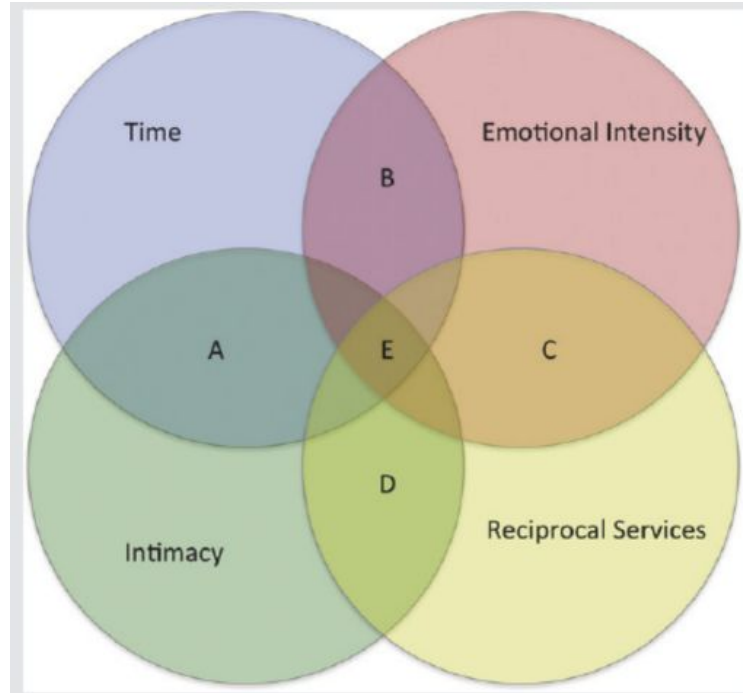
Tie Strength

- **Tie Strength** is the **measure of the strength of relationship** between the people.
- **Strong Ties** -rare , family members and close friends
 - Strong ties are **more trusted and their information is more likely to be reliable.**
- **Weak Ties** - includes acquaintances and more casual relationships.
 - Weak ties always **connect to diverse groups of people with different perspectives.**
- **Absent Ties** - The people with whom we have no relations.

Measuring tie strength-Granovetter's four principals

- **Time** can include the amount of time people spend with each other, the duration of their relationship (i.e., how long they have known each other), and how frequently they see one another.
- **Emotional intensity** is indicated by the closeness of a relationship; close friends or family members are likely to be strong ties, while more casual friends and acquaintances would be weaker ties.
- **Intimacy**-relates to **people sharing secrets** or intimate personal details with one another. The more of this information they exchange, the closer their relationship is likely to be.
- **Reciprocal Services** are favors that people do for one another. They may be personal (e.g., pet sitting or picking up someone's dry cleaning), financial (e.g., loaning money), professional (e.g., putting people in contact with one another), or otherwise.

Granovetter's four principals



Measuring tie strength -other factors

- **Structural features** relate to the social network of the two people in question. Those who have many mutual friends are likely to have stronger ties.
- **Social Distance measures** how different people's social situations are. This includes factors like age difference, race, education, and socioeconomic status. People with strong ties tend to have similar social attributes.
- **Emotional Support** describes the **communication between people** that validates their emotions, shows understanding of their problems, and tries to alleviate stress.

Measuring tie strength

- **Intimacy**
 - • Number of days since their last communication
 - • Number of friends in common
 - • Number of “intimate” words in their communications, as determined by software that automatically analyzes text
- **Intensity**
 - • Number of words exchanged on one another’s walls
 - • Depth of email threads in their inboxes (i.e., how many messages were sent back and forth in a conversation)
- **Reciprocal services**
 - • Number of links shared on one another’s wall
 - • Applications the users had in common (presumably because they could be working together within the application context)
- **Social distance**
 - • Age difference
 - • Difference in the number of educational degrees
 - • Difference in the number of occupations

Tie strength and network structure

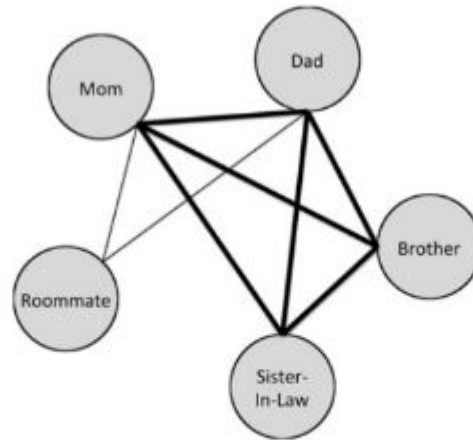


FIGURE 5.1

Sample Exercise. Note that there are strong ties connecting four of the five people listed, and two more weak ties. Only two ties are absent, between the roommate and brother, and roommate and sister-in-law. This is a very densely connected network with many strong ties.

Tie strength and network structure

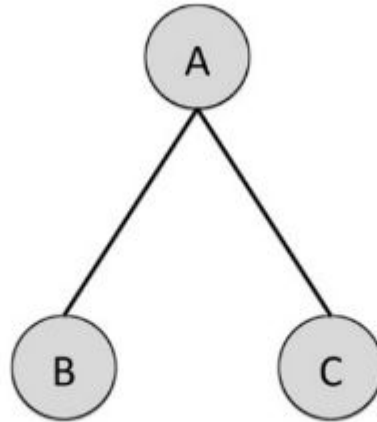


FIGURE 5.2

The Forbidden Triad.

When the tie does not exist it is called as forbidden triad

Tie strength and network structure

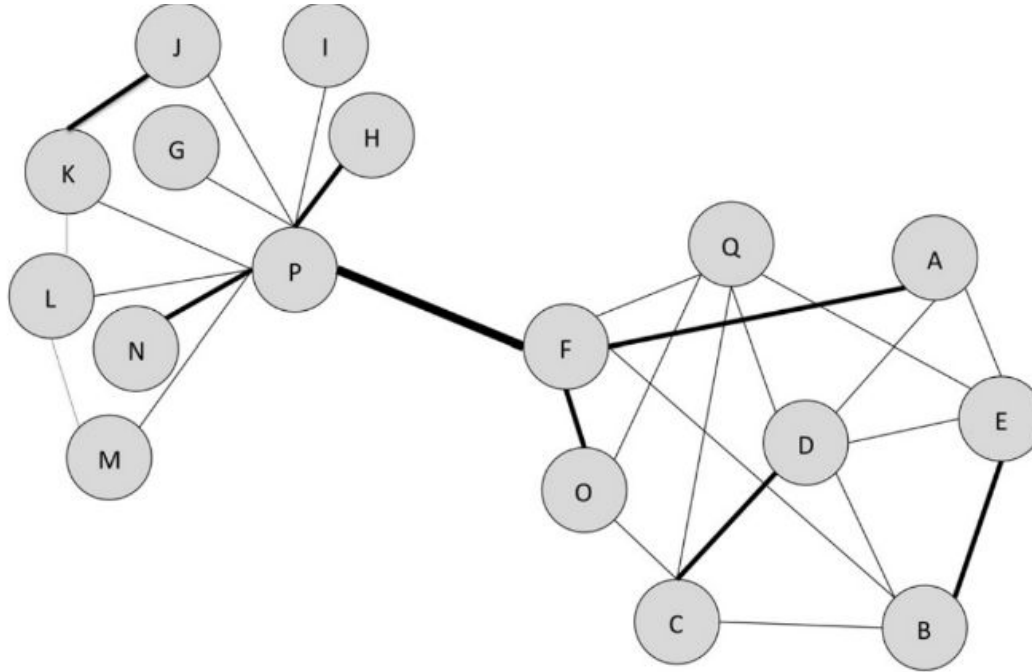


FIGURE 5.3

The edge between P and F is a bridge that connects the two clusters of nodes. It is a strong tie, and thus we would expect connections between some of the triads with two strong ties (e.g. PFO, PFH, PFN). It is very unlikely that no tie third tie would exist in any of those triads, and thus it is unlikely that a strong tie would be a bridge.

Network propagation

- Network propagation is a phenomenon where things spread through a network.
- How tie strength relates to the rate at which phenomena can spread through a network?

Network propagation

- Granovetter -study about finding jobs.
- A person has **more weak ties** than **strong ties**, and because the weak ties are connected to diverse social groups that the person would not otherwise communicate with, the **weak ties are more often sources of new information, like job opportunities.**

Tie strength and network propagation

- **Weak ties** are powerful in helping spread information farther through a network, but this is not to say that strong ties are unimportant.
- **Strong ties** tend to be more trustworthy, reliable, and personal.
- A weak tie may be able to tell a person about many job opportunities, but a strong tie will have a better idea of what jobs would be a good fit.
- **Strong ties** are also more likely to do things to help one another (i.e., reciprocal services).

Trust

Trust

- A person trusts another if she is willing to take a risk based on her expectation that the trusted person's actions will lead to a positive outcome.
- Trust is formed through a combination of factors such as **successful interactions, personal experiences, shared beliefs and values, and the presence of guarantees and protections** offered by institutions.

Trust

- **1. Calculation-based trust:** This is a rational decision about whether to trust someone, and where the costs and benefits of trusting are factored in.
- **2. Personal-based trust:** This reflects a person's propensity to trust, developed over the course of their life.
- **3. Cognition -based trust:** This describes the instant rapport and trust that can develop between people who share similar backgrounds, beliefs, and values. It often is based on first impressions.
- **4. Institution-based trust:** This addresses how trust may form in the presence of guarantees and protections offered by an institution.

Trust -Asymmetry

- Children must have almost absolute trust in their parents, while the parents may have almost no trust in their children, particularly when they are very young.
- **Context and time**
 - Someone may trust her friend to recommend a movie but not to repair her car.
 - People tend to **develop trust over time** as their history of shared experiences builds.
 - Trust may also decrease.
 - If someone has one dramatic failure, trust may disappear completely.

Measuring trust

- Decisions about trust in others deal with the risks a person is willing to take with a specific person.
 - 1. Trust with material possessions
 - 2. Belief about reliability
 - 3. Trust with secrets
 - 4. Trust regarding physical safety

Measuring trust

- **Trust in social media** is **harder to judge** compared to in-person interactions as the information available about a person is limited and identity can be easily forged.
- **Before social media**, users were mostly concerned with trusting websites, especially **e-commerce sites**, and retailers tried to overcome this by creating **privacy policies**, assuring transaction safety and security, and building professional and secure websites.
- With the rise of social media and user interaction, **trust issues shifted to include both websites and other users**.
- **Reputation systems**, such as the one **used by eBay**, were introduced to help users make decisions about **trust based on feedback from other users**.

Measuring trust

- **Social media users often share a lot of personal information**, some of which is private, and must trust both the website and other users to treat their information with respect.
- **Complex issues** regarding trust in social media arise due to the constantly changing factors of **what users want to share, with whom they want to share it, and what control they have over those decisions.**
- Reputation systems are not always present in social media applications and understanding trust in social media is a complex and constantly evolving topic.

Measuring trust

- The problem of inferring trust arises in social networks, where trust between two people can be considered a **weight on the edge connecting them**.
- **The determination of trust** is influenced by **an individual's propensity to trust and their assessment of the other person's trustworthiness**.
- Surveys could be used to generate a trust rating, but most people will not take the time to fill them out.

Network-based inference

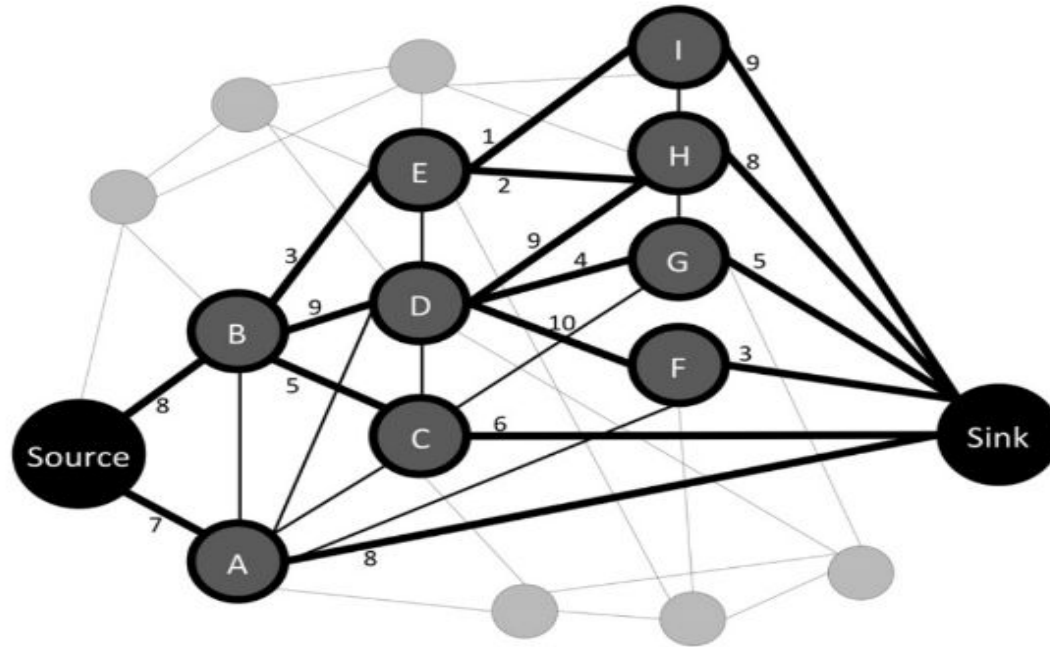


FIGURE 6.1

A social network with trust values shown as number weights on some of the edges. Trust is rated on a scale from 1 to 10 where 1 is low trust and 10 is high trust.

Network-based inference

1. The social network contains nodes for each person and edges with trust ratings (1 to 10).
2. The network can have multiple paths between the Source and Sink.
3. Some paths may have high trust values while others may have low trust values.
4. The Source can use **centrality measures to assess the trustworthiness** of the Sink based on the information in the social network.
5. One example of a centrality measure is the **weighted harmonic centrality**, which considers the weighted distances between a node and all other nodes in the network.
6. Another approach is to use the **PageRank algorithm**, which assigns a score to each node in the network based on the trust ratings of its neighbors.
7. The source would need to gather information about the trust ratings of all paths between the Source and the Sink.
8. Based on this information, the source can calculate centrality measures or PageRank scores to make a reasonably informed estimate of the Sink's trustworthiness.

Similarity-based trust inference

- People in social networks do not always supply trust ratings.
- Research shows that people who trust each other tend to be similar in some way.
- Similarity is important, but other factors such as major disagreements or disagreements on important issues can impact trust.
- **Trust can be estimated using similarities in item ratings**, - When people have provided ratings of items (e.g., they have rated movies), the similarities in their ratings can be used to estimate trust.
- **Estimating trust based on item ratings.**

Social Network Analytics

Common Social Media Networks

- **Friendship network**
- **Follow-Following Network**
- **FAN Network**
- **Group Network** - people who share common interests and agendas.eg. Yahoo Groups, Facebook Groups
- **Professional Networks** - Nodes in these networks are, for example, people, brands, and organizations, and links are professional relations (such as coworker, employee, or collaborator). An important feature of professional networks is the endorsement feature, where people who know you can endorse your skills and qualification.

Common Social Media Networks

- **Co authorship Network**
- **Hyperlink Network**
- **COLIKE** -colike networks are formed when two or more people like the same social media content.
- **COCOMMENTER NETWORKS**
- Cocommenter networks are formed when two or more people comment on socialmedia content (e.g., a Facebook status update, blog post, or YouTube video).
- Nodes represent users, and link represents the cocommenting relationship.

TYPES OF NETWORKS

- BASED ON EXISTENCE -

- **Implicit** (keyword co occurrence networks, co citation networks, cocommenter networks, hyperlink networks)
- **Explicit network-** (keyword cooccurrence networks, cocitation networks, cocommenter networks, hyperlink networks)

- BASED ON DIRECTION

- 1) directed networks
- 2) undirected networks.

TYPES OF NETWORKS

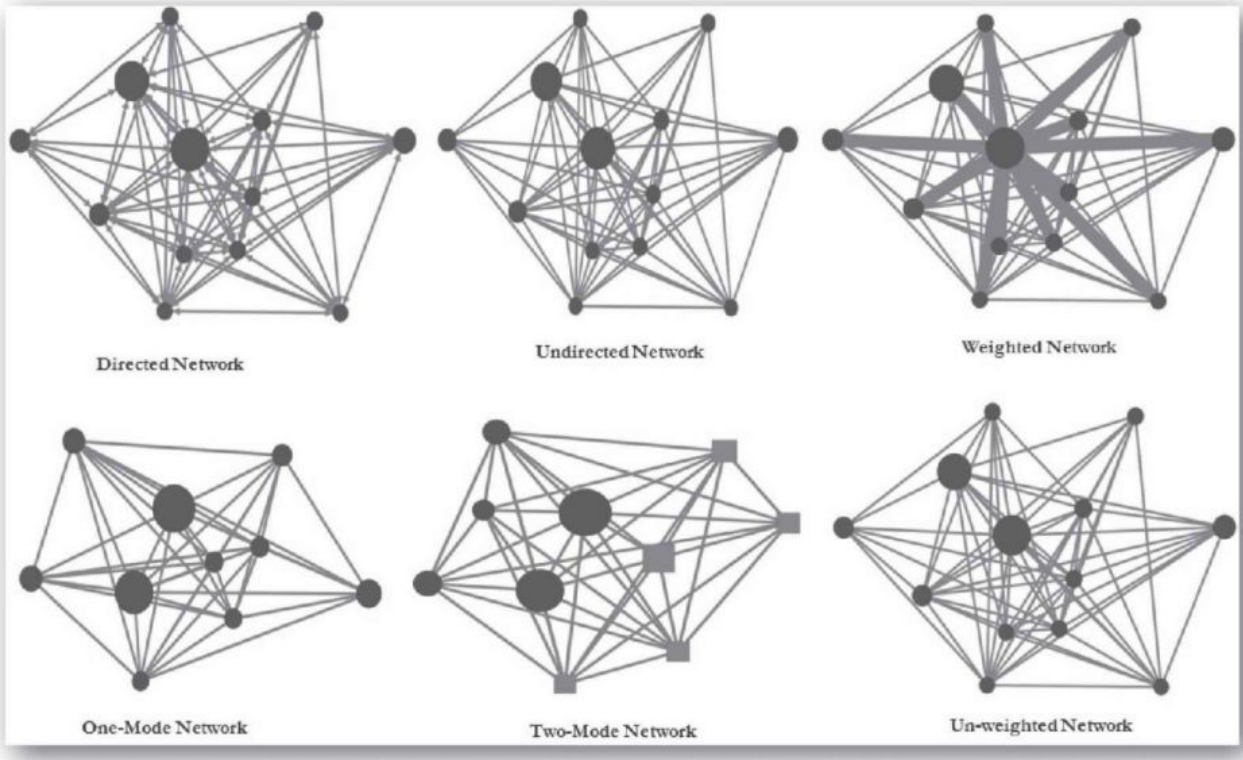
- BASED ON MODE

- Based on the composition of nodes, networks can be classified as
- 1) **One-mode network** - A one-mode network is formed among a single set of nodes of the same nature . A Facebook friendship network is an example of a one-mode network where nodes (people) form network ties (friendships).
- 2) **Two-mode networks** - Two-mode networks (also known as bipartite networks) are networks with two sets of nodes of different classes.one set of nodes (circles) could be social media users and other set of nodes (squares) could be participation in a series of events. Users are linked to the events they attended.
- 3) **Multimode networks**.

- BASED ON WEIGHTS

- 1) weighted networks,
- 2) unweighted networks.

TYPES OF NETWORKS



COMMON NETWORK TERMINOLOGIES

NODE-LEVEL PROPERTIES

- Degree Centrality
- Betweenness Centrality
- Eigenvector Centrality

NETWORK-LEVEL PROPERTIES

- **Clustering Coefficient** - The clustering coefficient of a network is the degree to which nodes in a network tend to cluster or group together.
- **Density** -Density can be calculated as the number of links present in a network divided by the number of all possible links between pairs of nodes in a network
- **Components** -Components of a network are the isolated sub-networks that connect within, but are disconnected between, sub-networks.
- **Diameter** -The diameter of a network is the largest of all the calculated shortest path between any pair of nodes in a network
- **Average Degree** -The average degree centrality measures the average number of links among nodes in a network.

NETWORK ANALYTICS TOOLS

- NodeXL: A free tool for social network analysis and visualization for networks on platforms like Facebook, Twitter, and YouTube.
- UCINET: A social network analysis software for Windows operating system with a free 90-day trial.
- Pajek: A software application for analyzing and visualizing large networks, free for non-commercial use on Windows operating systems.
- Netminer: A software application for large social network analysis and visualization, with a free 28-day trial.
- Flocker: A real-time Twitter analytics tool for retweets and mentions networks.
- Reach: An online platform for mapping hashtag networks and identifying the most influential accounts on Twitter.
- Mentionmapp: An online tool for investigating Twitter mentions networks.

NETWORK ANALYTICS TOOLS - NodeXL

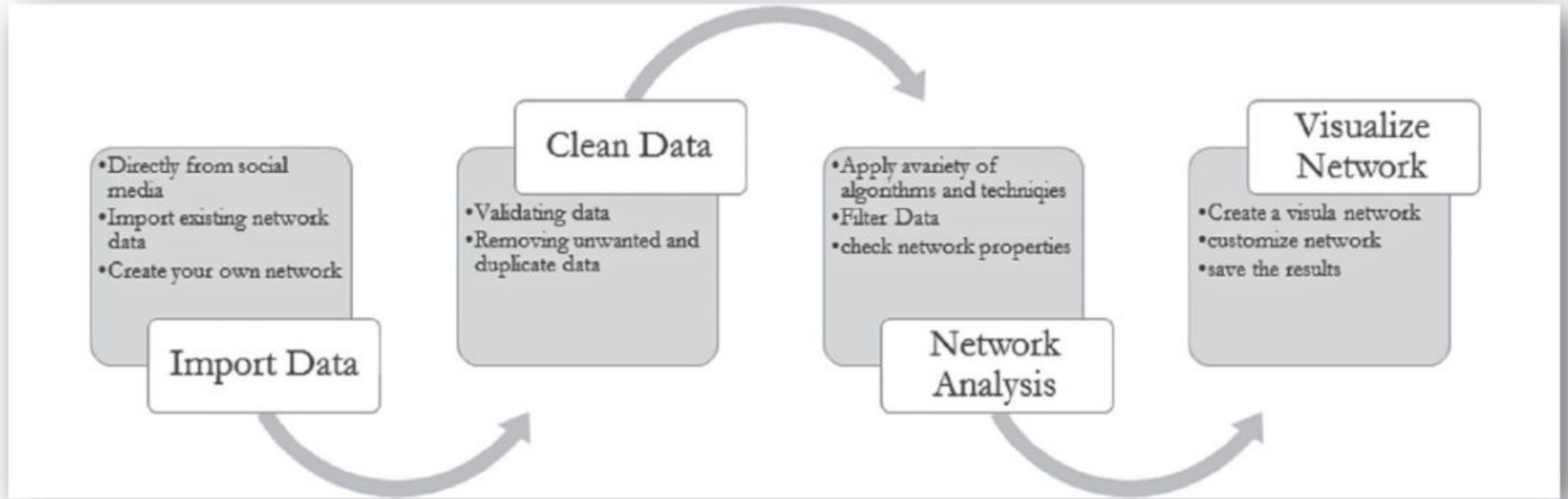
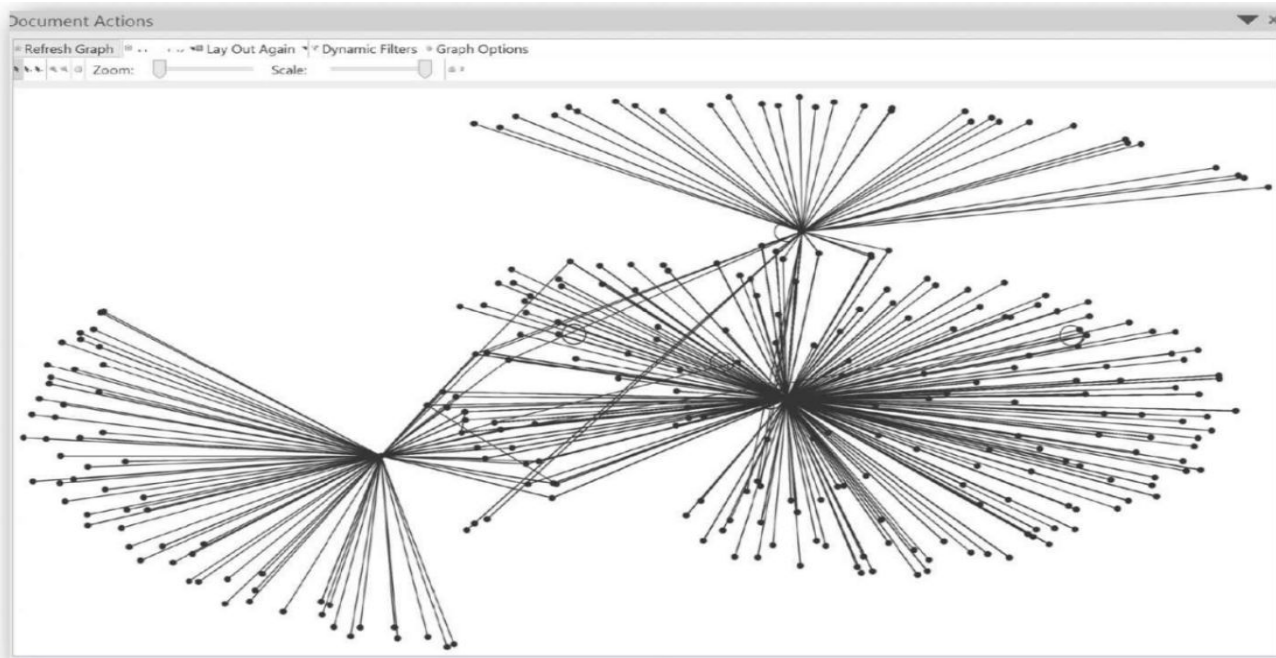


Figure 8. NodeXL work flow stages

NETWORK ANALYTICS TOOLS - NodeXL



Screenshot 8. Facebook fan page network

Explain the Granovetter's four principles of measuring tie strength

Questions

- 1. What is a network?
- 2. What is the purpose of network analytics?
- 3. Briefly differentiate among social networks, social network sites, social networking, and social network analysis.
- 4. Briefly explain the different types of social media networks.
- 5. What is the difference between explicit and implicit networks?
- 6. What is the difference between one-mode, two-mode, and multimode networks?
- 7. Differentiate between weighted and unweighted networks.
- 8. Briefly define important node level properties, such as degree, betweenness, eigenvector centralities, and structural holes.
- 9. Briefly explain important network-level properties, such as clustering coefficient, density, diameter, average degree, and components.
- Explain the importance of Weak Ties and Strong ties
- What is tie strength and how to measure it?
- Explain the Granovetter's four principles of measuring tie strength